# Author Rebuttal for Submission ICML 2026

Authors

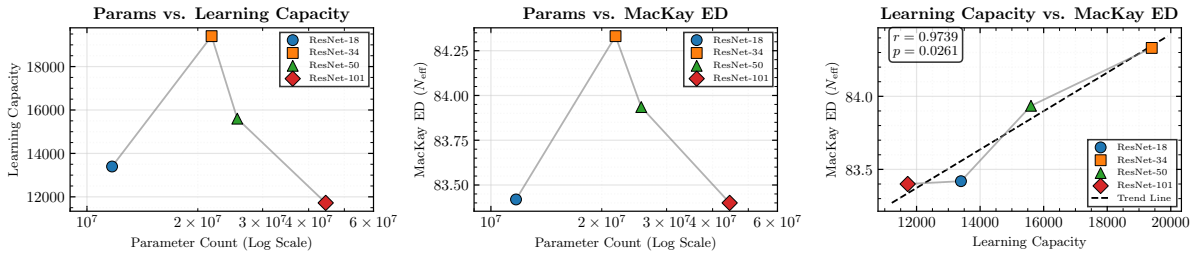


Figure 1: Empirical Validation: Learning Capacity vs. MacKay ED on ResNet.

Table 1: Empirical Validation: Learning Capacity vs. MacKay ED on ResNet (CIFAR-100).

| **Architecture** | **Params (M)** | **Learning Capacity** | **MacKay ED ($N_{\text{eff}}$)** |
|---|---|---|---|
| ResNet-18 | 11.7 | 13,397.186 | 83.4193 |
| ResNet-34 | 21.8 | 19,400.860 | 84.3313 |
| ResNet-50 | 25.6 | 15,601.120 | 83.9353 |
| ResNet-101 | 44.5 | 11,721.313 | 83.4001 |
| | | *Pearson correlation (r):* | **0.9739** |
| | | *p-value:* | **0.0261** |

Table 2: Robustness Ratios (*Adv/Clean*) for the MIM Attack across budgets $\epsilon$. Architecture families: **ShuffleNetV2** (SN) and **RepVGG** (RV). $r$ denotes the Pearson correlation coefficient between $N_{eff}$ and the robustness ratio.

| Fam. | Model | $N_{eff} \downarrow$ | $\epsilon = \frac{1}{255}$ | $\epsilon = \frac{2}{255}$ | $\epsilon = \frac{4}{255}$ | $\epsilon = \frac{8}{255}$ | $\epsilon = \frac{16}{255}$ |
|---|---|---|---|---|---|---|---|
| | **CIFAR-10** | | | | | | |
| RV | `repvgg_a1` | 79.2 | 0.518 | 0.448 | 0.347 | 0.240 | 0.210 |
| | `repvgg_a0` | 78.3 | 0.514 | 0.442 | 0.344 | 0.232 | 0.202 |
| | `repvgg_a2` | 78.0 | 0.535 | 0.462 | 0.354 | 0.239 | 0.207 |
| | Pearson coefficient ($r$) | | -0.58 | -0.45 | -0.45 | 0.36 | 0.67 |
| SN | `shufflenetv2_x0_5` | 81.6 | 0.428 | 0.345 | 0.233 | 0.135 | 0.110 |
| | `shufflenetv2_x1_0` | 78.1 | 0.482 | 0.403 | 0.285 | 0.157 | 0.124 |
| | `shufflenetv2_x1_5` | 78.0 | 0.470 | 0.390 | 0.270 | 0.159 | 0.127 |
| | `shufflenetv2_x2_0` | 77.6 | 0.481 | 0.402 | 0.289 | 0.174 | 0.142 |
| | Pearson coefficient ($r$) | | -0.97 | -0.98 | -0.96 | -0.94 | -0.86 |
| | **CIFAR-100** | | | | | | |
| RV | `repvgg_a0` | 87.4 | 0.294 | 0.225 | 0.143 | 0.088 | 0.071 |
| | `repvgg_a1` | 86.7 | 0.305 | 0.231 | 0.158 | 0.093 | 0.076 |
| | `repvgg_a2` | 84.4 | 0.313 | 0.251 | 0.172 | 0.103 | 0.086 |
| | Pearson coefficient ($r$) | | -0.93 | -1.00 | -0.95 | -0.99 | -1.00 |
| SN | `shufflenetv2_x0_5` | 88.0 | 0.263 | 0.194 | 0.117 | 0.062 | 0.049 |
| | `shufflenetv2_x1_0` | 87.5 | 0.283 | 0.209 | 0.125 | 0.072 | 0.056 |
| | `shufflenetv2_x1_5` | 87.0 | 0.281 | 0.209 | 0.124 | 0.070 | 0.057 |
| | `shufflenetv2_x2_0` | 86.0 | 0.282 | 0.221 | 0.150 | 0.084 | 0.070 |
| | Pearson coefficient ($r$) | | -0.68 | -0.93 | -0.95 | -0.95 | -0.97 |

## MIM Attack

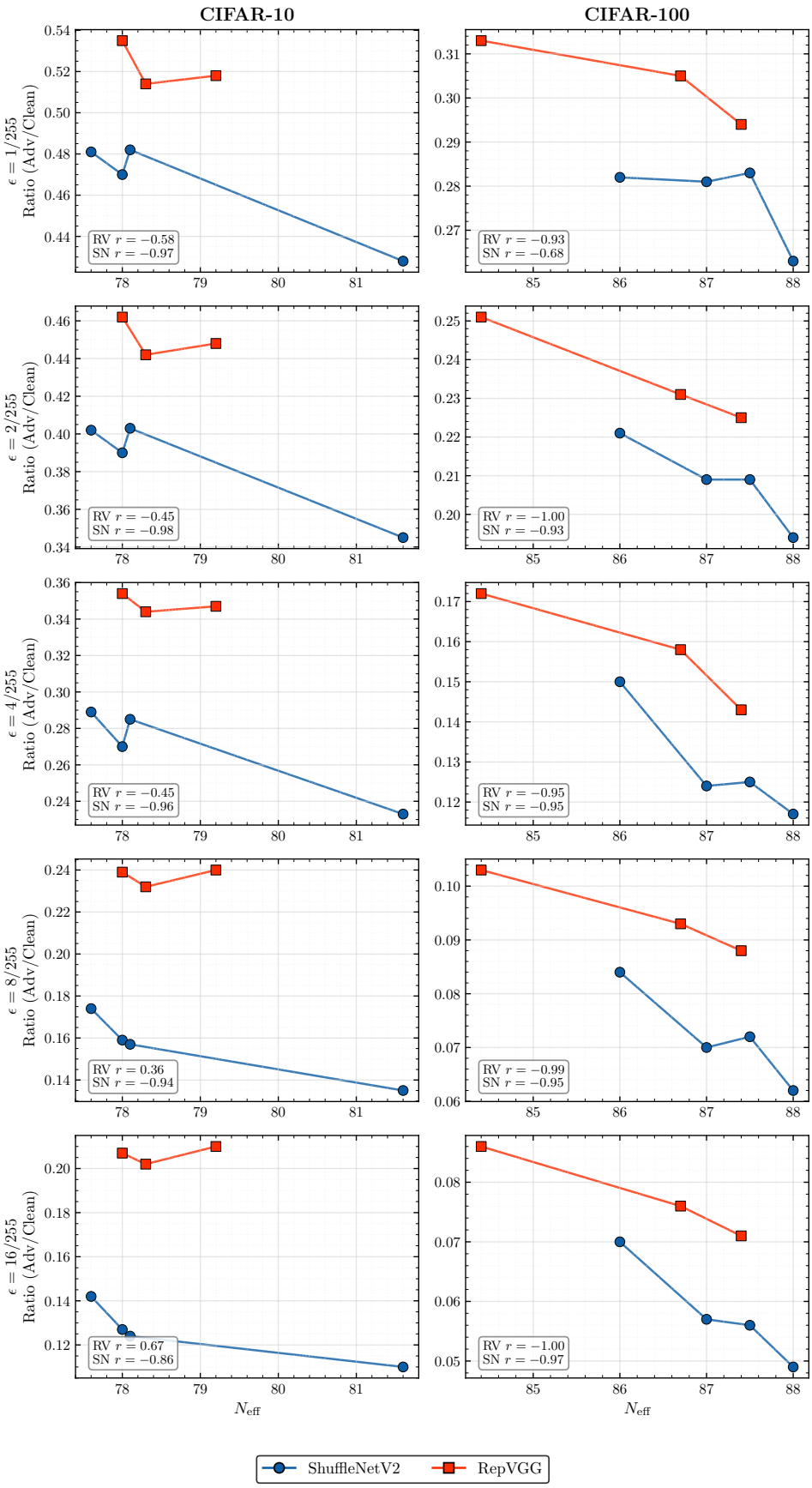


Figure 2: Robustness Ratios for the MIM Attack across budgets $\epsilon$.

Table 3: Robustness Ratios (*Adv/Clean*) for the Square Attack across budgets $\epsilon$. Architecture families: **ShuffleNetV2** (SN) and **RepVGG** (RV). $r$ denotes the Pearson correlation coefficient between $N_{eff}$ and the robustness ratio.

| Fam. | Model | $N_{eff} \downarrow$ | $\epsilon = \frac{1}{255}$ | $\epsilon = \frac{2}{255}$ | $\epsilon = \frac{4}{255}$ | $\epsilon = \frac{8}{255}$ | $\epsilon = \frac{16}{255}$ |
|---|---|---|---|---|---|---|---|
| | **CIFAR-10** | | | | | | |
| RV | `repvgg_a1` | 79.2 | 0.557 | 0.544 | 0.520 | 0.470 | 0.378 |
| | `repvgg_a0` | 78.3 | 0.553 | 0.542 | 0.516 | 0.463 | 0.367 |
| | `repvgg_a2` | 78.0 | 0.572 | 0.560 | 0.537 | 0.487 | 0.386 |
| | Pearson coefficient ($r$) | | -0.54 | -0.61 | -0.55 | -0.44 | -0.16 |
| SN | `shufflenetv2_x0_5` | 81.6 | 0.479 | 0.462 | 0.428 | 0.365 | 0.264 |
| | `shufflenetv2_x1_0` | 78.1 | 0.533 | 0.515 | 0.482 | 0.425 | 0.319 |
| | `shufflenetv2_x1_5` | 78.0 | 0.520 | 0.503 | 0.474 | 0.415 | 0.311 |
| | `shufflenetv2_x2_0` | 77.6 | 0.526 | 0.511 | 0.482 | 0.425 | 0.323 |
| | Pearson coefficient ($r$) | | -0.96 | -0.97 | -0.99 | -0.99 | -0.99 |
| | **CIFAR-100** | | | | | | |
| RV | `repvgg_a0` | 87.4 | 0.345 | 0.331 | 0.300 | 0.245 | 0.154 |
| | `repvgg_a1` | 86.7 | 0.361 | 0.345 | 0.313 | 0.255 | 0.163 |
| | `repvgg_a2` | 84.4 | 0.352 | 0.340 | 0.316 | 0.266 | 0.175 |
| | Pearson coefficient ($r$) | | -0.17 | -0.40 | -0.77 | -0.96 | -0.97 |
| SN | `shufflenetv2_x0_5` | 88.0 | 0.309 | 0.293 | 0.263 | 0.213 | 0.137 |
| | `shufflenetv2_x1_0` | 87.5 | 0.329 | 0.316 | 0.285 | 0.228 | 0.142 |
| | `shufflenetv2_x1_5` | 87.0 | 0.326 | 0.310 | 0.282 | 0.230 | 0.141 |
| | `shufflenetv2_x2_0` | 86.0 | 0.327 | 0.315 | 0.287 | 0.237 | 0.156 |
| | Pearson coefficient ($r$) | | -0.63 | -0.68 | -0.76 | -0.90 | -0.94 |

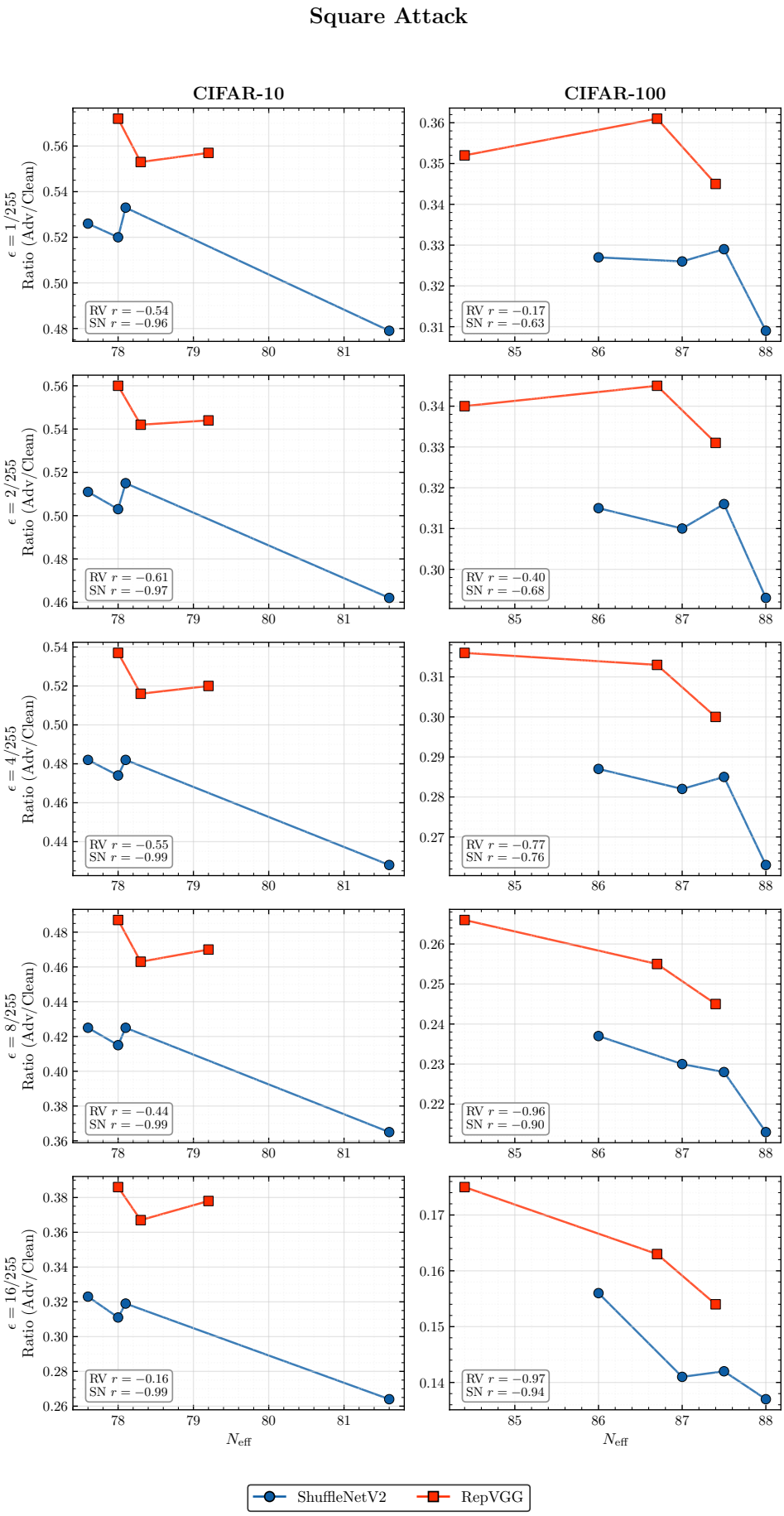


Figure 3: Robustness Ratios for the Square Attack across budgets $\epsilon$.

Table 4: Mean Robustness Ratios ($Adv/Clean$) $\pm$ std for PGD-10 across budgets $\epsilon$. Architecture families: **ShuffleNetV2** (SN) and **RepVGG** (RV). $r$ denotes the Pearson correlation coefficient.

| Fam. | Model | $N_{eff} \downarrow$ | $\epsilon = \frac{1}{255}$ | $\epsilon = \frac{2}{255}$ | $\epsilon = \frac{4}{255}$ | $\epsilon = \frac{8}{255}$ | $\epsilon = \frac{16}{255}$ |
|---|---|---|---|---|---|---|---|
| | | | **CIFAR-10** | | | | |
| RV | `repvgg_a0` | $78.1 \pm 0.12$ | $0.519 \pm 0.005$ | $0.440 \pm 0.006$ | $0.310 \pm 0.003$ | $0.155 \pm 0.005$ | $0.085 \pm 0.005$ |
| | `repvgg_a1` | $78.6 \pm 0.15$ | $0.518 \pm 0.005$ | $0.439 \pm 0.007$ | $0.308 \pm 0.012$ | $0.155 \pm 0.008$ | $0.084 \pm 0.004$ |
| | `repvgg_a2` | $78.4 \pm 0.10$ | $0.525 \pm 0.006$ | $0.446 \pm 0.004$ | $0.309 \pm 0.008$ | $0.157 \pm 0.004$ | $0.083 \pm 0.002$ |
| | Pearson coefficient ($r$) | | -0.06 | 0.06 | -0.97 | 0.04 | -0.67 |
| RV | `repvgg_b0` | $79.8 \pm 0.22$ | $0.566 \pm 0.021$ | $0.508 \pm 0.017$ | $0.407 \pm 0.013$ | $0.280 \pm 0.000$ | $0.187 \pm 0.004$ |
| | `repvgg_b1` | $78.8 \pm 0.19$ | $0.572 \pm 0.002$ | $0.520 \pm 0.001$ | $0.422 \pm 0.002$ | $0.282 \pm 0.000$ | $0.192 \pm 0.003$ |
| | `repvgg_b2` | $79.0 \pm 0.10$ | $0.571 \pm 0.003$ | $0.521 \pm 0.003$ | $0.419 \pm 0.004$ | $0.280 \pm 0.001$ | $0.196 \pm 0.002$ |
| | Pearson coefficient ($r$) | | -1.00 | -0.99 | -1.00 | -0.61 | -0.80 |
| SN | `shufflenetv2_x0_5` | $81.8 \pm 0.42$ | $0.453 \pm 0.019$ | $0.352 \pm 0.014$ | $0.208 \pm 0.004$ | $0.081 \pm 0.005$ | $0.035 \pm 0.002$ |
| | `shufflenetv2_x1_0` | $78.4 \pm 0.35$ | $0.474 \pm 0.005$ | $0.384 \pm 0.006$ | $0.240 \pm 0.004$ | $0.097 \pm 0.005$ | $0.042 \pm 0.002$ |
| | `shufflenetv2_x1_5` | $77.5 \pm 0.42$ | $0.475 \pm 0.010$ | $0.385 \pm 0.007$ | $0.244 \pm 0.004$ | $0.100 \pm 0.004$ | $0.045 \pm 0.003$ |
| | `shufflenetv2_x2_0` | $77.0 \pm 0.43$ | $0.493 \pm 0.010$ | $0.405 \pm 0.012$ | $0.261 \pm 0.004$ | $0.113 \pm 0.006$ | $0.050 \pm 0.003$ |
| | Pearson coefficient ($r$) | | -0.94 | -0.96 | -0.97 | -0.95 | -0.96 |
| | | | **CIFAR-100** | | | | |
| RV | `repvgg_a0` | $87.3 \pm 0.12$ | $0.298 \pm 0.004$ | $0.216 \pm 0.004$ | $0.121 \pm 0.003$ | $0.052 \pm 0.003$ | $0.024 \pm 0.003$ |
| | `repvgg_a1` | $86.7 \pm 0.08$ | $0.310 \pm 0.004$ | $0.232 \pm 0.007$ | $0.133 \pm 0.004$ | $0.057 \pm 0.001$ | $0.028 \pm 0.001$ |
| | `repvgg_a2` | $85.3 \pm 0.42$ | $0.313 \pm 0.002$ | $0.241 \pm 0.002$ | $0.142 \pm 0.002$ | $0.067 \pm 0.003$ | $0.031 \pm 0.001$ |
| | Pearson coefficient ($r$) | | -0.83 | -0.91 | -0.92 | -0.99 | -0.96 |
| RV | `repvgg_b0` | $82.1 \pm 0.25$ | $0.351 \pm 0.004$ | $0.320 \pm 0.005$ | $0.208 \pm 0.007$ | $0.120 \pm 0.003$ | $0.087 \pm 0.002$ |
| | `repvgg_b1` | $87.4 \pm 0.20$ | $0.302 \pm 0.007$ | $0.291 \pm 0.010$ | $0.202 \pm 0.009$ | $0.112 \pm 0.002$ | $0.047 \pm 0.002$ |
| | `repvgg_b2` | $87.1 \pm 0.32$ | $0.343 \pm 0.002$ | $0.261 \pm 0.001$ | $0.201 \pm 0.001$ | $0.111 \pm 0.002$ | $0.085 \pm 0.000$ |
| | Pearson coefficient ($r$) | | -0.67 | -0.83 | -0.99 | -0.98 | -0.58 |
| SN | `shufflenetv2_x0_5` | $88.1 \pm 0.13$ | $0.260 \pm 0.017$ | $0.179 \pm 0.014$ | $0.091 \pm 0.008$ | $0.035 \pm 0.003$ | $0.017 \pm 0.001$ |
| | `shufflenetv2_x1_0` | $87.6 \pm 0.37$ | $0.281 \pm 0.009$ | $0.197 \pm 0.006$ | $0.104 \pm 0.003$ | $0.042 \pm 0.001$ | $0.019 \pm 0.000$ |
| | `shufflenetv2_x1_5` | $87.0 \pm 0.05$ | $0.274 \pm 0.008$ | $0.197 \pm 0.002$ | $0.105 \pm 0.005$ | $0.043 \pm 0.002$ | $0.019 \pm 0.001$ |
| | `shufflenetv2_x2_0` | $86.0 \pm 0.03$ | $0.293 \pm 0.012$ | $0.219 \pm 0.010$ | $0.125 \pm 0.007$ | $0.056 \pm 0.004$ | $0.026 \pm 0.001$ |
| | Pearson coefficient ($r$) | | -0.86 | -0.96 | -0.97 | -0.98 | -0.97 |

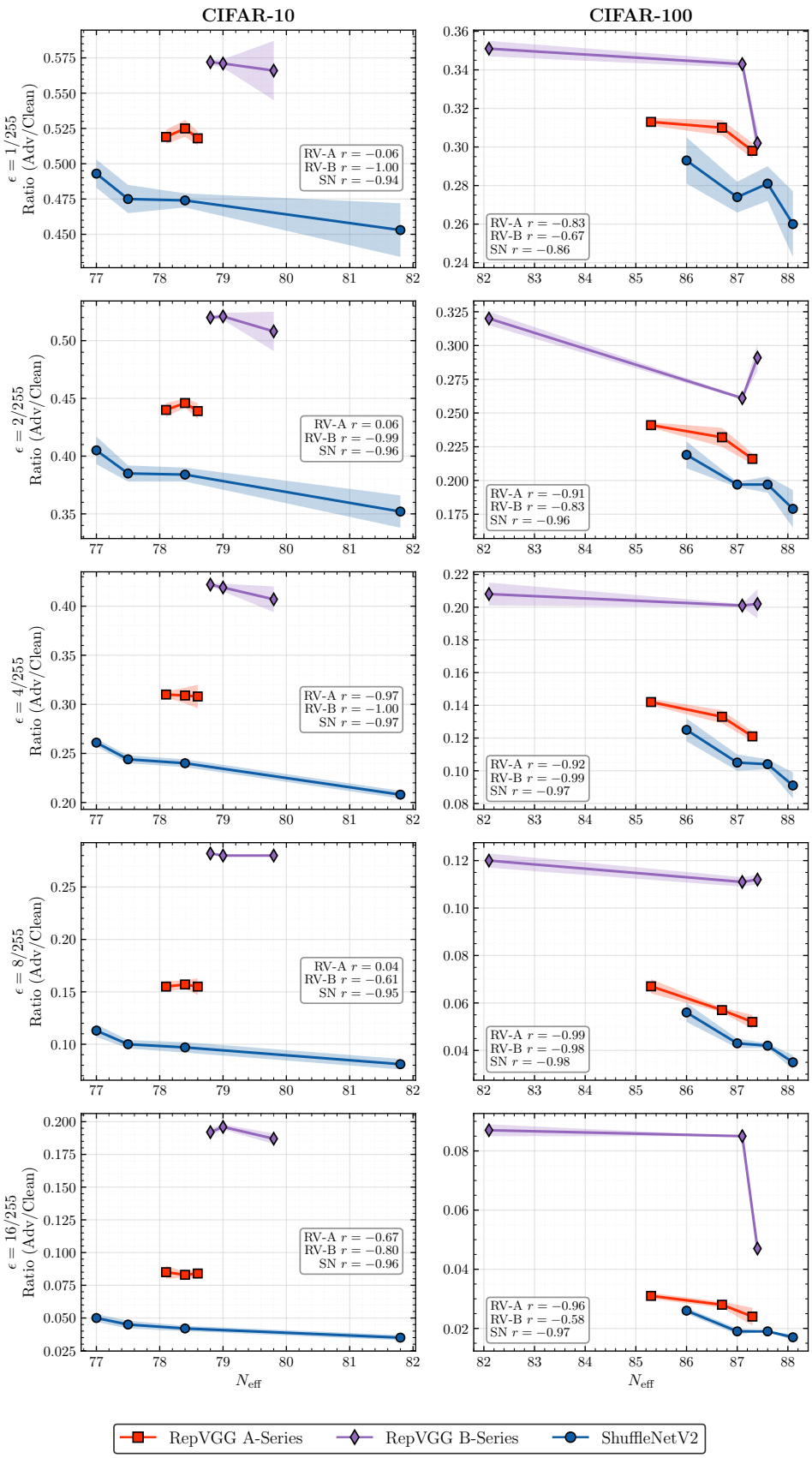


Figure 4: PGD-10 Attack: Variance Envelopes showing Mean Robustness.

## Proof of Relation Between Learning Capacity and MacKay's ED Under Bounded Local Integration

### The Eigenspace-Aligned Bounded Laplace Approximation

For a model with parameters $w \in R^k$, sample size $N$, and regularized loss function $\hat{L}(w) = NL(w) + \frac{\epsilon}{2}\|w\|^2$, the partition function assuming a uniform prior is initially given by the integral over the entire weight space:

$$Z(N) = \int e^{-\hat{L}(w)}\, dw \tag{1}$$

In highly non-convex deep neural networks, the Laplace approximation is mathematically accurate only within the immediate quadratic basin of attraction around a converged local minimum $w^*$. To formalize this, we restrict the integration domain to a localized bounded volume $\mathcal{B}$ centered at $w^*$.

To evaluate this integral while preserving the independence of the principal curvatures, we adopt the spectral projection methodology established by Chen et al. [2024] (see Appendix F.2, Eq. 34 therein). Specifically, we define the boundary $\mathcal{B}$ as a hyper-rectangle aligned perfectly with the orthogonal eigenvectors of the local Hessian. Let the bound in each principal direction be $[-B, B]$, where $B$ represents the finite radius of the local basin along the axes of curvature.

Performing a second-order Taylor expansion around the local minimum $w^*$ (where the gradient $\nabla\hat{L}(w^*) = 0$) and substituting the Hessian $H = N\nabla^2 L(w^*) + \epsilon I$, the bounded partition function becomes:

$$Z(N) \approx e^{-\hat{L}(w^*)} \int_{\mathcal{B}} \exp\left(-\frac{1}{2}(w - w^*)^T H (w - w^*)\right) dw \tag{2}$$

### Spectral Separation and the Gauss Error Function

We transform the integral into the orthogonal eigenspace of $H$. Let $x = Q(w - w^*)$, where $Q$ is the orthogonal matrix of eigenvectors such that $H = Q^T \Lambda Q$. The eigenvalues of $H$ are $\alpha_i = N\lambda_i + \epsilon$, which populate the diagonal matrix $\Lambda$.

Because the domain $\mathcal{B}$ is explicitly defined along these eigenvectors following the projection approach of Chen et al. [2024], the cross-terms are eliminated and the $k$-dimensional integral strictly separates into a product of $k$ independent 1D bounded Gaussian integrals:

$$Z(N) \approx e^{-\hat{L}(w^*)} \prod_{i=1}^{k} \int_{-B}^{B} \exp\left(-\frac{1}{2}\alpha_i x_i^2\right) dx_i \tag{3}$$

Using the standard evaluation for the bounded Gaussian integral, $\int_{-B}^{B} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}\,\mathrm{erf}(B\sqrt{a})$, where $\mathrm{erf}(\cdot)$ is the Gauss error function, we obtain:

$$Z(N) \approx e^{-\hat{L}(w^*)} \prod_{i=1}^{k} \left[\sqrt{\frac{2\pi}{\alpha_i}}\,\mathrm{erf}\left(B\sqrt{\frac{\alpha_i}{2}}\right)\right] \tag{4}$$

Taking the logarithm yields the free energy for the bounded local basin. Let $z_i = B\sqrt{\frac{\alpha_i}{2}} = B\sqrt{\frac{N\lambda_i + \epsilon}{2}}$ represent the scaled boundary limit in the $i$-th dimension:

$$\log Z(N) \approx -\hat{L}(w^*) + \frac{k}{2}\log(2\pi) + \sum_{i=1}^{k}\left[-\frac{1}{2}\log(\alpha_i) + \log\mathrm{erf}(z_i)\right] \tag{5}$$

### Derivation of Bounded Learning Capacity

Learning Capacity is defined formally by Chen et al. [2024] via the continuous second derivative as $C(N) = N^2 \frac{\partial^2}{\partial N^2}\log Z(N)$. We differentiate the spectral sum in Eq. (5) with respect to $N$. Let $g(\alpha_i) = -\frac{1}{2}\log(\alpha_i) + \log\mathrm{erf}(z_i)$.

**First Derivative:** Using the chain rule, $\frac{\partial z_i}{\partial \alpha_i} = \frac{z_i}{2\alpha_i}$, and the derivative of the error function $\frac{d}{dz} \log \operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \frac{e^{-z^2}}{\operatorname{erf}(z)}$, we find:

$$g'(\alpha_i) = -\frac{1}{2\alpha_i} + \left( \frac{2}{\sqrt{\pi}} \frac{e^{-z_i^2}}{\operatorname{erf}(z_i)} \right) \frac{z_i}{2\alpha_i} = -\frac{1}{2\alpha_i}(1 - \delta_i) \tag{6}$$

where we define the geometric attenuation factor as $\delta_i = \frac{2z_i e^{-z_i^2}}{\sqrt{\pi} \operatorname{erf}(z_i)}$.

**Second Derivative:** To differentiate $g'(\alpha_i)$ with respect to $\alpha_i$, we first require the derivative of $\delta_i$. Taking the logarithmic derivative of $\delta_i$ with respect to $z_i$ explicitly yields:

$$\frac{d\delta_i}{dz_i} = \frac{\delta_i}{z_i}(1 - 2z_i^2 - \delta_i) \tag{7}$$

Applying the chain rule $\frac{\partial \delta_i}{\partial \alpha_i} = \frac{d\delta_i}{dz_i} \frac{\partial z_i}{\partial \alpha_i}$, we get:

$$\frac{\partial \delta_i}{\partial \alpha_i} = \frac{\delta_i}{2\alpha_i}(1 - 2z_i^2 - \delta_i) \tag{8}$$

Substituting this into the derivative of $g'(\alpha_i)$ and factoring out $\frac{1}{2\alpha_i^2}$ yields:

$$g''(\alpha_i) = \frac{1}{2\alpha_i^2}(1 - \delta_i) + \frac{1}{2\alpha_i} \left[ \frac{\delta_i}{2\alpha_i}(1 - 2z_i^2 - \delta_i) \right] = \frac{1}{2\alpha_i^2} \left[ 1 - \delta_i + \frac{\delta_i}{2}(1 - 2z_i^2 - \delta_i) \right] \tag{9}$$

We condense the boundary correction term as $\Phi_i = \frac{\delta_i}{2}(1 + 2z_i^2 + \delta_i)$.

Applying the chain rule for $N$ (where $\frac{\partial \alpha_i}{\partial N} = \lambda_i$), the second derivative of the partition function becomes:

$$\frac{\partial^2}{\partial N^2} \log Z(N) \approx \sum_{i=1}^{k} \lambda_i^2 g''(\alpha_i) = \frac{1}{2} \sum_{i=1}^{k} \frac{\lambda_i^2}{(N\lambda_i + \epsilon)^2}(1 - \Phi_i) \tag{10}$$

**Calculating Learning Capacity:** Multiplying by $N^2$, we arrive at the final analytical form for the Learning Capacity strictly within a bounded local domain:

$$C_{\text{bounded}}(N) = \frac{1}{2} \sum_{i=1}^{k} \left( \frac{N\lambda_i}{N\lambda_i + \epsilon} \right)^2 (1 - \Phi_i) \tag{11}$$

**Relationship with MacKay's Effective Dimensionality**

Using the spectral component $\gamma_i = \frac{\lambda_i}{\lambda_i + \epsilon/N}$ (the "determinedness" of the $i$-th parameter direction), MacKay's Effective Dimensionality is defined as $N_{\text{eff}} = \sum_{i=1}^{k} \gamma_i$. Noting that $\frac{N\lambda_i}{N\lambda_i + \epsilon}$ is mathematically equivalent to $\gamma_i$, the relationship under the strictly bounded local Laplace approximation becomes:

$$C_{\text{bounded}}(N) = \frac{1}{2} \sum_{i=1}^{k} \gamma_i^2 (1 - \Phi_i) \tag{12}$$

This derivation establishes that restricting the functional space to a localized hypercube around a converged minimum strictly *attenuates* the Learning Capacity. The term $\Phi_i = \frac{\delta_i}{2}(1 + 2z_i^2 + \delta_i) > 0$ acts as a precise geometric penalty reflecting the truncated volume of the local basin.

Crucially, as a theoretical limit to prove generalization, if the radius of the basin expands ($B \to \infty$), the scaled boundary $z_i \to \infty$. In this limit, $\operatorname{erf}(z_i) \to 1$ and the exponential term $e^{-z_i^2} \to 0$ exponentially fast. This causes the boundary decay rate $\delta_i \to 0$, and consequently, the geometric penalty $\Phi_i \to 0$. Thus, the boundary correction naturally vanishes, perfectly recovering the continuous unbounded relationship $C(N) = \frac{1}{2} \sum \gamma_i^2$ as a rigorous upper bound.